

$$\begin{aligned}
& (g_1^2 + g_2^2) + \\
& \left(\frac{3}{8} g_1^4 + g_1^2 g_2^2 + \frac{23}{24} g_2^4 + \frac{4}{9} C_{\phi 2^2} g_2^4 L_{F3,0}[m_2] + \frac{2}{3} C_{\phi 2^2} g_2^4 L_{F4,-1}[m_2] - \frac{32}{45} C_{\phi 2^2} g_2^4 L_{F5,-2}[m_2] - \right. \\
& \frac{1}{12} \sum_{\mathbf{p}} g_1^4 L_{F2,0}[m_{\mathbf{d}}^{\mathbf{p}}] + g_1^2 \overline{y_{\mathbf{d}}^{\mathbf{p}}} y_{\mathbf{d}}^{\mathbf{p}} L_{F2,0}[m_{\mathbf{d}}^{\mathbf{r}}] - \frac{1}{4} \sum_{\mathbf{p}} g_1^4 L_{F2,0}[m_{\mathbf{l}}^{\mathbf{p}}] + \\
& g_1^2 \overline{y_{\mathbf{e}}^{\mathbf{p}}} y_{\mathbf{e}}^{\mathbf{p}} L_{F2,0}[m_{\mathbf{e}}^{\mathbf{r}}] + \frac{1}{8} (4 \overline{y_{\mathbf{e}}^{\mathbf{p}}} y_{\mathbf{e}}^{\mathbf{p}} (-g_1^2 + g_2^2) - \sum_{\mathbf{p}} (g_1^4 + g_2^4)) L_{F2,0}[m_{\mathbf{l}}^{\mathbf{p}}] + \frac{1}{18} C_{\phi 2^2} \\
& g_2^2 (36 \overline{y_{\mathbf{e}}^{\mathbf{p}}} y_{\mathbf{e}}^{\mathbf{p}} - 5 \sum_{\mathbf{p}} g_2^2) L_{F3,0}[m_{\mathbf{l}}^{\mathbf{p}}] + \frac{1}{12} C_{\phi 2^2} g_2^2 (-24 \overline{y_{\mathbf{e}}^{\mathbf{p}}} y_{\mathbf{e}}^{\mathbf{p}} + \sum_{\mathbf{p}} g_2^2) L_{F4,-1}[m_{\mathbf{l}}^{\mathbf{p}}] + \\
& \frac{8}{45} \sum_{\mathbf{p}} C_{\phi 2^2} g_2^4 L_{F5,-2}[m_{\mathbf{l}}^{\mathbf{p}}] + \left(\frac{1}{2} \overline{y_{\mathbf{d}}^{\mathbf{p}}} y_{\mathbf{d}}^{\mathbf{p}} (g_1^2 + 3 g_2^2) - \frac{1}{24} \sum_{\mathbf{p}} (g_1^4 + 9 g_2^4) \right) L_{F2,0}[m_{\mathbf{q}}^{\mathbf{p}}] + \\
& \frac{1}{6} C_{\phi 2^2} g_2^2 (36 \overline{y_{\mathbf{d}}^{\mathbf{p}}} y_{\mathbf{d}}^{\mathbf{p}} - 5 \sum_{\mathbf{p}} g_2^2) L_{F3,0}[m_{\mathbf{q}}^{\mathbf{p}}] + \frac{1}{4} C_{\phi 2^2} g_2^2 (-24 \overline{y_{\mathbf{d}}^{\mathbf{p}}} y_{\mathbf{d}}^{\mathbf{p}} + \sum_{\mathbf{p}} g_2^2) L_{F4,-1}[m_{\mathbf{q}}^{\mathbf{p}}] + \\
& \frac{8}{15} \sum_{\mathbf{p}} C_{\phi 2^2} g_2^4 L_{F5,-2}[m_{\mathbf{q}}^{\mathbf{p}}] - \frac{1}{3} \sum_{\mathbf{p}} g_1^4 L_{F2,0}[m_{\mathbf{u}}^{\mathbf{p}}] + \frac{2}{9} C_{\phi 2^2} g_2^4 L_{F3,0}[\tilde{\mu}] + \\
& \frac{1}{3} C_{\phi 2^2} g_2^4 L_{F4,-1}[\tilde{\mu}] - \frac{16}{45} C_{\phi 2^2} g_2^4 L_{F5,-2}[\tilde{\mu}] + \frac{1}{2} g_1^4 L_{F2,2,-2}[m_1, \tilde{\mu}] + \\
& \frac{1}{2} g_1^4 m_1^2 L_{F2,2,-1}[m_1, \tilde{\mu}] - \frac{8}{3} C_{\phi 2^2} g_2^4 L_{F2,1,0}[m_2, \tilde{\mu}] + \frac{5}{2} g_2^4 L_{F2,2,-2}[m_2, \tilde{\mu}] + \\
& \frac{1}{6} g_2^4 (-34 C_{\phi 2^2} + 3 m_2^2) L_{F2,2,-1}[m_2, \tilde{\mu}] + \frac{2}{3} m_2 \tilde{\mu} g_2^4 (\overline{C_{\phi 1\phi 2}} + C_{\phi 1\phi 2}) L_{F2,2,0}[m_2, \tilde{\mu}] + \\
& \frac{4}{3} C_{\phi 2^2} g_2^4 L_{F3,1,-1}[m_2, \tilde{\mu}] + \frac{28}{3} C_{\phi 2^2} g_2^4 L_{F3,2,-2}[m_2, \tilde{\mu}] - \\
& \frac{1}{3} m_2 g_2^4 (18 m_2 C_{\phi 2^2} + \tilde{\mu} (\overline{C_{\phi 1\phi 2}} + C_{\phi 1\phi 2})) L_{F3,2,-1}[m_2, \tilde{\mu}] - 4 C_{\phi 2^2} g_2^4 L_{F3,3,-3}[m_2, \tilde{\mu}] + \\
& 4 C_{\phi 2^2} g_2^4 m_2^2 L_{F3,3,-2}[m_2, \tilde{\mu}] - 4 C_{\phi 2^2} g_2^4 L_{F4,2,-3}[m_2, \tilde{\mu}] + 4 C_{\phi 2^2} g_2^4 m_2^2 L_{F4,2,-2}[m_2, \tilde{\mu}] - \\
& 3 \overline{y_{\mathbf{d}}^{\mathbf{p}}} \overline{y_{\mathbf{d}}^{\mathbf{t}}} y_{\mathbf{d}}^{\mathbf{p}} y_{\mathbf{d}}^{\mathbf{t}} L_{F1,1,0}[m_{\mathbf{d}}^{\mathbf{r}}, m_{\mathbf{d}}^{\mathbf{t}}] + g_1^2 \overline{a_{\mathbf{d}}^{\mathbf{p}}} a_{\mathbf{d}}^{\mathbf{p}} L_{F2,1,0}[m_{\mathbf{d}}^{\mathbf{r}}, m_{\mathbf{q}}^{\mathbf{p}}] + \\
& \frac{1}{2} g_1^2 (\overline{C_{\phi 1\phi 2}} \tilde{\mu} \overline{y_{\mathbf{d}}^{\mathbf{p}}} a_{\mathbf{d}}^{\mathbf{p}} + \overline{a_{\mathbf{d}}^{\mathbf{p}}} (2 C_{\phi 2^2} a_{\mathbf{d}}^{\mathbf{p}} + C_{\phi 1\phi 2} \tilde{\mu} y_{\mathbf{d}}^{\mathbf{p}})) L_{F2,2,0}[m_{\mathbf{d}}^{\mathbf{r}}, m_{\mathbf{q}}^{\mathbf{p}}] - \\
& \overline{y_{\mathbf{e}}^{\mathbf{p}}} \overline{y_{\mathbf{e}}^{\mathbf{t}}} y_{\mathbf{e}}^{\mathbf{p}} y_{\mathbf{e}}^{\mathbf{t}} L_{F1,1,0}[m_{\mathbf{e}}^{\mathbf{r}}, m_{\mathbf{e}}^{\mathbf{t}}] + g_1^2 \overline{a_{\mathbf{e}}^{\mathbf{p}}} a_{\mathbf{e}}^{\mathbf{p}} L_{F2,1,0}[m_{\mathbf{e}}^{\mathbf{r}}, m_{\mathbf{l}}^{\mathbf{p}}] + \\
& \frac{1}{2} g_1^2 (\overline{C_{\phi 1\phi 2}} \tilde{\mu} \overline{y_{\mathbf{e}}^{\mathbf{p}}} a_{\mathbf{e}}^{\mathbf{p}} + \overline{a_{\mathbf{e}}^{\mathbf{p}}} (2 C_{\phi 2^2} a_{\mathbf{e}}^{\mathbf{p}} + C_{\phi 1\phi 2} \tilde{\mu} y_{\mathbf{e}}^{\mathbf{p}})) L_{F2,2,0}[m_{\mathbf{e}}^{\mathbf{r}}, m_{\mathbf{l}}^{\mathbf{p}}] + \\
& g_2^2 \overline{a_{\mathbf{e}}^{\mathbf{p}}} a_{\mathbf{e}}^{\mathbf{p}} L_{F2,1,0}[m_{\mathbf{l}}^{\mathbf{p}}, m_{\mathbf{e}}^{\mathbf{r}}] - \frac{1}{2} \overline{a_{\mathbf{e}}^{\mathbf{p}}} a_{\mathbf{e}}^{\mathbf{p}} (g_1^2 + g_2^2) L_{F3,1,-1}[m_{\mathbf{l}}^{\mathbf{p}}, m_{\mathbf{e}}^{\mathbf{r}}] + \\
& g_2^2 (\overline{C_{\phi 1\phi 2}} \tilde{\mu} \overline{y_{\mathbf{e}}^{\mathbf{p}}} a_{\mathbf{e}}^{\mathbf{p}} + \overline{a_{\mathbf{e}}^{\mathbf{p}}} (2 C_{\phi 2^2} a_{\mathbf{e}}^{\mathbf{p}} + C_{\phi 1\phi 2} \tilde{\mu} y_{\mathbf{e}}^{\mathbf{p}})) L_{F3,1,0}[m_{\mathbf{l}}^{\mathbf{p}}, m_{\mathbf{e}}^{\mathbf{r}}] - \\
& \frac{1}{2} g_1^2 (\overline{C_{\phi 1\phi 2}} \tilde{\mu} \overline{y_{\mathbf{e}}^{\mathbf{p}}} a_{\mathbf{e}}^{\mathbf{p}} + \overline{a_{\mathbf{e}}^{\mathbf{p}}} (2 C_{\phi 2^2} a_{\mathbf{e}}^{\mathbf{p}} + C_{\phi 1\phi 2} \tilde{\mu} y_{\mathbf{e}}^{\mathbf{p}})) L_{F3,2,-1}[m_{\mathbf{l}}^{\mathbf{p}}, m_{\mathbf{e}}^{\mathbf{r}}] + \\
& \frac{1}{4} (-2 C_{\phi 2^2} \overline{a_{\mathbf{e}}^{\mathbf{p}}} a_{\mathbf{e}}^{\mathbf{p}} (3 g_1^2 + 5 g_2^2) - \tilde{\mu} (3 g_1^2 + 7 g_2^2) (\overline{C_{\phi 1\phi 2}} \overline{y_{\mathbf{e}}^{\mathbf{p}}} a_{\mathbf{e}}^{\mathbf{p}} + C_{\phi 1\phi 2} y_{\mathbf{e}}^{\mathbf{p}} \overline{a_{\mathbf{e}}^{\mathbf{p}}})) \\
& L_{F4,1,-1}[m_{\mathbf{l}}^{\mathbf{p}}, m_{\mathbf{e}}^{\mathbf{r}}] + \\
& \frac{1}{3} (2 C_{\phi 2^2} \overline{a_{\mathbf{e}}^{\mathbf{p}}} a_{\mathbf{e}}^{\mathbf{p}} (3 g_1^2 + g_2^2) + \tilde{\mu} (3 g_1^2 + 2 g_2^2) (\overline{C_{\phi 1\phi 2}} \overline{y_{\mathbf{e}}^{\mathbf{p}}} a_{\mathbf{e}}^{\mathbf{p}} + C_{\phi 1\phi 2} y_{\mathbf{e}}^{\mathbf{p}} \overline{a_{\mathbf{e}}^{\mathbf{p}}})) \\
& L_{F5,1,-2}[m_{\mathbf{l}}^{\mathbf{p}}, m_{\mathbf{e}}^{\mathbf{r}}] - \overline{y_{\mathbf{e}}^{\mathbf{p}}} \overline{y_{\mathbf{e}}^{\mathbf{t}}} y_{\mathbf{e}}^{\mathbf{p}} y_{\mathbf{e}}^{\mathbf{t}} L_{F1,1,0}[m_{\mathbf{l}}^{\mathbf{p}}, m_{\mathbf{l}}^{\mathbf{s}}] - \\
& 2 C_{\phi 2^2} \overline{y_{\mathbf{e}}^{\mathbf{p}}} \overline{y_{\mathbf{e}}^{\mathbf{t}}} y_{\mathbf{e}}^{\mathbf{p}} y_{\mathbf{e}}^{\mathbf{t}} L_{F2,1,0}[m_{\mathbf{l}}^{\mathbf{p}}, m_{\mathbf{l}}^{\mathbf{s}}] + 2 C_{\phi 2^2} \overline{y_{\mathbf{e}}^{\mathbf{p}}} \overline{y_{\mathbf{e}}^{\mathbf{t}}} y_{\mathbf{e}}^{\mathbf{p}} y_{\mathbf{e}}^{\mathbf{t}} L_{F3,1,-1}[m_{\mathbf{l}}^{\mathbf{p}}, m_{\mathbf{l}}^{\mathbf{s}}] + \\
& \overline{a_{\mathbf{d}}^{\mathbf{p}}} a_{\mathbf{d}}^{\mathbf{p}} (2 g_1^2 + 3 g_2^2) L_{F$$